

CGS 698C: Assignment 1

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1.1

(a) A sample space Ω , is the set of all possible outcomes of a random experiment. When two coins are tossed, each coin can land as either 'Heads (H)' or 'Tails (T)'.
 So, $\Omega = \{HH, HT, TH, TT\}$

(b) An event space is the power set of sample space, or the set of all possible subsets of sample space.
 If $n(\Omega) = K$, then $n(P(\Omega)) = 2^K$
 Here, $K = 4 \Rightarrow n(P(\Omega)) = 2^4 = 16$ possible events.

i) Empty set (or, Impossible event) = $\emptyset$

(ii) Elementary events = $\{HH\}, \{HT\}, \{TH\}, \{TT\}$

(iii) Two outcomes = $\{HH, HT\}, \{HH, TH\}, \{HH, TT\},$
 $\{HT, TH\}, \{HT, TT\}, \{TH, TT\}$

(iv) Three outcomes = $\{HH, HT, TH\}, \{HH, HT, TT\}, \{HH, TH, TT\},$
 $\{HT, TH, TT\}$

(v) Four outcomes = $\{HH, HT, TH, TT\}$

(c) Given that, $P(\{HH\}) = P(\{HT\}) = P(\{TH\}) = P(\{TT\}) = K$ (let)

From the second and third axiom of probability.

We know that, $\sum_{x \in \Omega} f(x) = 1$

where, f assigns a probability to each elementary event of x .

So, $P(\{HH\}) + P(\{HT\}) + P(\{TH\}) + P(\{TT\}) = 1$

$\Rightarrow K + K + K + K = 1$

$$\Rightarrow 4K = 1 \Rightarrow K = \frac{1}{4}$$

$\therefore$ Probability of occurrence of each elementary event is $\frac{1}{4}$

(ii) Let A be the event that at least one head ~~occurs~~ appears, then its possible outcomes are: $\{HH, HT, TH\}$

Now, $P(A) = P(\{HH\}) + P(\{HT\}) + P(\{TH\})$

$\therefore P(A) = P(\{HH\} \cup \{HT\} \cup \{TH\})$

and using third axiom we get

$$P(A) = P(\{HH\}) + P(\{HT\}) + P(\{TH\})$$

$$= \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4} \quad \therefore P(A) = \frac{3}{4}$$

$\therefore$ Probability of occurrence of at least one heads is $\frac{3}{4}$

(iii) B be the event, exactly one head appears, outcomes are $\{HT, TH\}$

Using third axiom,

$$P(B) = P(\{HT\}) + P(\{TH\}) = \frac{1}{4} + \frac{1}{4}$$

$$\therefore P(B) = \frac{1}{2}$$

2.1

The probability assigner function is,

$$f(n, k, p) = \binom{n}{k} p^k (1-p)^{n-k}$$

where, k is number of correctly recognised words, n is total words and p is the underlying probability of correctly recognising a word.

Given: $n = 50$, $k = 45$, $p = 0.9$

$$\begin{aligned} f(50, 45, 0.9) &= \binom{50}{45} (0.9)^{45} (0.1)^5 \\ &= \frac{50!}{45! 5!} \times (0.9)^{45} \times (0.1)^5 \\ &\approx 0.185 \quad (\text{approx.}) \end{aligned}$$

2.2. probability mass function is,

$$f(k, \lambda) = \frac{\lambda^k}{k!} e^{-\lambda}$$

where, $\lambda = 10$

(i) probability that zero road accidents in a day,
 $k=0 \Rightarrow f(0, 10) = \frac{10^0}{0!} e^{-10} = \underline{\underline{e^{-10}}}$

(ii) probability of more than 7 but less than 10 road accidents in a day $= P(8 \leq k \leq 9)$

$$\begin{aligned} \text{Now, } P(8 \leq k \leq 9) &= f(8, 10) + f(9, 10) \\ &= \frac{10^8}{8!} e^{-10} + \frac{10^9}{9!} e^{-10} \end{aligned}$$

$$\therefore \boxed{P(8 \leq k \leq 9) = \frac{10^8}{9!} \times 19 \times e^{-10}}$$

(iii) Graph of Probability mass function :

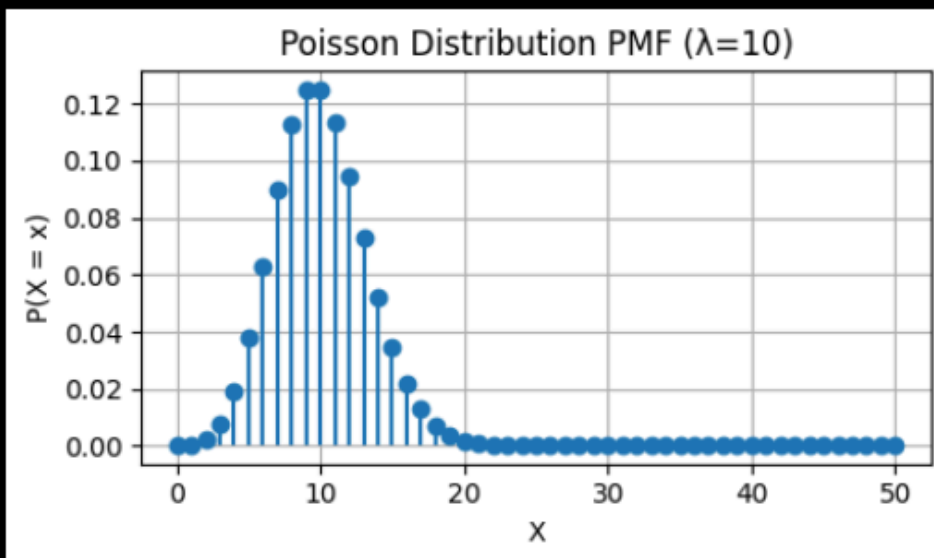
```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import poisson
```

[1] ✓ 14.4s

```
λ = 10
x = np.arange(0, 51)
pmf_values = poisson.pmf(x, mu=λ)

plt.figure(figsize=(5, 3))
plt.stem(x, pmf_values, basefmt=" ")
plt.title(f"Poisson Distribution PMF (λ={λ})")
plt.xlabel("X")
plt.ylabel("P(X = x)")
plt.grid(True)
plt.tight_layout()
plt.show()
```

[4] ✓ 0.1s



3.1 Probability Density function of normal distribution is given by $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

(i) find p.d.f for $x=0$, given $\mu=1$, $\sigma=1$

$$\Rightarrow f(0) = \frac{1}{1 \cdot \sqrt{2\pi}} e^{-\frac{(0-1)^2}{2 \cdot 1^2}}$$

$$\therefore \boxed{f(0) = \frac{1}{\sqrt{2\pi}e}}$$

(ii) find p.d.f for $x=1$, give $\mu=0$, $\sigma=1$

$$\Rightarrow f(1) = \frac{1}{1 \cdot \sqrt{2\pi}} e^{-\frac{1^2}{2 \cdot 1^2}} \Rightarrow \boxed{f(1) = \frac{1}{\sqrt{2\pi}e}}$$

(iii) Given, $P(x_1 \leq X \leq x_2) = 0.30$
 $P(x_1 \leq X \leq x_3) = 0.45$

Find, $P(x_2 \leq X \leq x_3) = ?$

$$\Rightarrow P(x_2 \leq X \leq x_3) = P(x_1 \leq X \leq x_3) - P(x_1 \leq X \leq x_2)$$
$$= 0.45 - 0.30$$

$$\therefore \boxed{P(x_2 \leq X \leq x_3) = 0.15}$$

4.1)

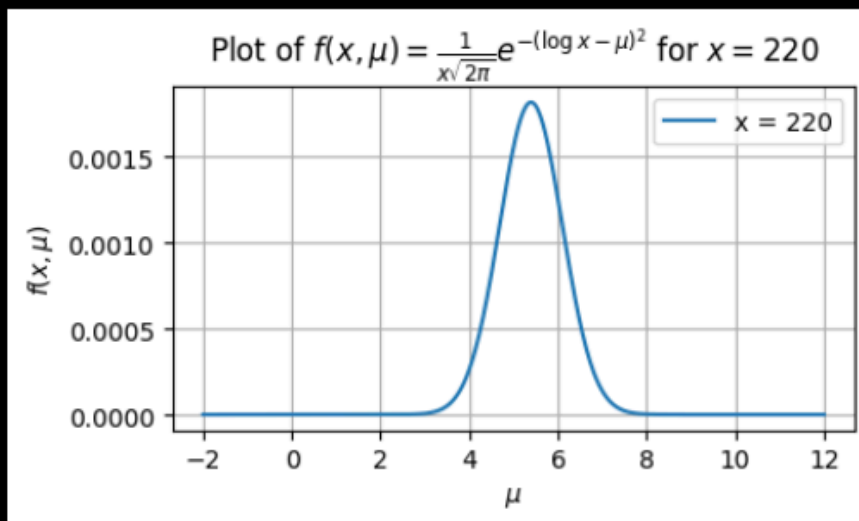
(a)

```
x = 220
mu_values = np.linspace(-2, 12, 500)

prefactor = 1 / (x * np.sqrt(2 * np.pi))
log_x = np.log(x)
f_values = prefactor * np.exp(-(log_x - mu_values)**2)

plt.figure(figsize=(5, 3))
plt.plot(mu_values, f_values, label=f'x = {x}')
plt.title(rf'Plot of $f(x, \mu) = \frac{1}{x\sqrt{2\pi}} e^{\{-(\log x - \mu)^2\}}$ for $x = \{x\}$')
plt.xlabel(r'$\mu$')
plt.ylabel(r'$f(x, \mu)$')
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()
```

✓ 0.2s



(b)

```
x_obs = np.array([303.25, 443, 220, 560, 880])
n = len(x_obs)
mu_values = np.linspace(-2, 10, 500)

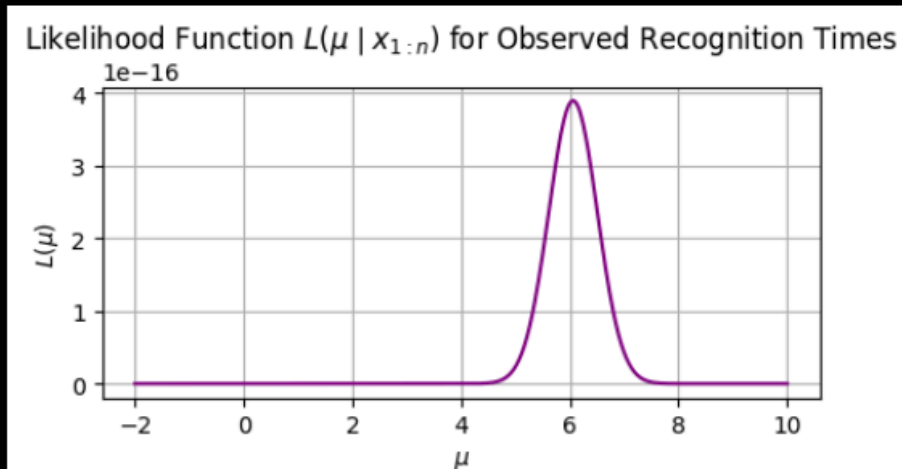
prefactor = 1 / (np.prod(x_obs) * (np.sqrt(2 * np.pi) ** n))
log_x = np.log(x_obs)

likelihoods = []
for mu in mu_values:
    exponent = -np.sum((log_x - mu) ** 2) / 2
    likelihood = prefactor * np.exp(exponent)
    likelihoods.append(likelihood)

likelihoods = np.array(likelihoods)

plt.figure(figsize=(5, 3))
plt.plot(mu_values, likelihoods, color='purple')
plt.title(r'Likelihood Function  $L(\mu | x_{1:n})$  for Observed Recognition Times')
plt.xlabel(r' $\mu$ ')
plt.ylabel(r' $L(\mu)$ ')
plt.grid(True)
plt.tight_layout()
plt.show()
```

✓ 0.1s



(c) Clearly, for $\mu = 6$ (approximately), the likelihood of obtaining the observed sample 303, 443, 220, 560, 880 is maximum. To be precise we can find the mean of $\log(x_i)$ which will be 6.0.619